

# Comparative Study on the Development Paths of U.S. and Soviet Cooperative Engagement Systems

## Comparative Study on the Development Paths of U.S. and Soviet Cooperative Engagement Systems

*A Counterfactual Deduction Centered on the “A-Shoots-B-Guides” Architecture*

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**Note.** This is a scholarly counterfactual study based entirely on publicly available sources. The descriptions of system configurations in 2026 given in Chapter 6 are the results of logical deduction under explicitly stated constraints; they are not statements of fact, intelligence assessments, or predictions. All materials cited herein come from public channels, and this work does not represent the position of any institution.

**Translator’s note on the colloquialisms “A-shoots-B-guides” and “A-locks-B-shoots-C-guides.”** Neither is standard military terminology in any language; both are popular Chinese-language shorthand discussed in Section 1.2. Note that the letters are reassigned between the two phrases, which can confuse readers unfamiliar with the source usage. In **“A-shoots-B-guides”** (A 射 B 导), **A** denotes the *firing* platform and **B** the *guiding* platform. In **“A-locks-B-shoots-C-guides”** (A 锁 B 射 C 导), **A** denotes the *detecting and locking* platform, **B** the *firing* platform, and **C** the *guiding* platform. This work treats the two phrases as adjacent levels on a single capability spectrum rather than as two distinct schemes, and preserves the original letter assignments throughout so that cross-references to the Chinese-language literature remain traceable. The corresponding formal U.S. terminology is *engage on remote*, implemented through Cooperative Engagement Capability (CEC) and Naval Integrated Fire Control–Counter Air (NIFC-CA).

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## Abstract

This study examines the U.S. and Soviet capabilities for separating detection from firepower, compares the divergent development paths of the two cooperative engagement systems under identical technological conditions, and uses counterfactual deduction to reconstruct the half of that comparison which was cut

short by the dissolution of the Soviet Union in 1991. Taking the colloquial expression “A-shoots-B-guides” as its entry point, the study decomposes it into three diagnostic indicators — whether the data link carries fire-control-quality raw data or processed tracks, whether the engagement loop is complete, and whether terminal guidance requires continuous illumination by a third party — and constructs a four-layer comparative framework covering sensors, data links, weapons, and command and control. The study finds that in 1991 the gap between the two sides in system conception was far smaller than the standard narrative suggests: at the architectural level, the Soviet air defense system already possessed the skeleton of detection–firepower separation, and its weaknesses lay in data precision and latency across command echelons rather than in the system concept itself. The American advantage was concentrated in a single engineering capability — fusing geographically dispersed radar data into a single fire-control-quality track picture. Setting three constraints (technological inheritance chains, engineering cycles, and the continuity of system design philosophy), the study deduces the system configurations both sides would have reached by 2026 had the Cold War continued, and cross-checks the results in two directions against the actual degradation of the Russian system after the Soviet collapse: the links the deduction judged to be blocked were indeed left unresolved for decades, and the evolutionary directions the deduction identified broadly match the technological choices Russia subsequently made. On this basis, system design logic constitutes an independent variable in explaining the outcome of this technological competition, and is not equivalent to the level of technological capability. The conclusion tests a counter-intuitive proposition: where both sides retain a second-strike capability and judge each other’s away-theater disadvantages accurately, the simultaneous maturation of cooperative engagement capability between two superpowers may have a net effect of inhibiting rather than encouraging preemption. The conditions under which that proposition ceases to hold are specified.

**Keywords:** Cooperative Engagement Capability; A-shoots-B-guides; counterfactual deduction; offense–defense balance; U.S.–Soviet military competition

## Chapter 1 · Introduction

### 1.1 Statement of the Problem

“A-shoots-B-guides” is not a standard military term. It is a popular formulation used by Chinese military media and enthusiasts for a mode of operations in which the platform that detects and locks onto a target is separate from the platform that launches the weapon, and the launching platform completes the shot on the basis of externally supplied data received over a data link while its own radar is either switched off or unable to see the target. The corresponding formal expression within the U.S. military is “engage on remote,” and the systems that implement it are Cooperative Engagement Capability and Naval Integrated Fire Control–Counter Air. “A-locks-B-shoots-C-guides” goes one step further on this basis, assigning detection

and locking, weapon launch, and midcourse guidance to three different platforms, with the missile transitioning to autonomous homing in its terminal phase [1].

Both concepts have attracted a great deal of discussion in China in recent years, but that discussion has mostly concentrated on the American side, or on cataloguing equipment performance parameters. The question that genuinely merits pursuit is a different one: at the end of the Cold War, the United States and the Soviet Union faced the same set of technological conditions — phased-array radar had matured, digital data links had become practical, and air-to-air and surface-to-air missiles were beginning to be fitted with active radar seekers. Why, then, did the two sides ultimately follow paths that differed so greatly?

The answer given by the U.S. Navy was the Cooperative Engagement Capability (CEC) system. CEC fuses the radar returns of all ships and aircraft within a formation into a single unified composite track picture, enabling platforms that have not themselves detected a target to fire on the basis of network data; together with the Aegis system, the E-2D airborne early warning aircraft, and the Standard Missile-6, it constitutes Naval Integrated Fire Control-Counter Air (NIFC-CA) [2]. The Soviet answer was entirely different. The Soviet air defense system was never designed on the principle that “every platform can fight independently”; it was designed on the principle that “the higher echelon sees the whole picture, the lower echelon does the shooting”: automated command and control systems connected multiple air defense systems into a single command chain, airborne early warning aircraft passed target coordinates down to ground-based air defense systems over data links, and combat units were permitted to deploy far from the guidance radar site [3]. In form, this architecture already possessed the skeleton of detection–firepower separation, but what it transmitted was processed track information, not fire-control-quality raw sensor data.

The Soviet Union dissolved in 1991, and this comparison was forcibly interrupted. Russia inherited the entire system, only to see it degrade continuously over the following three decades and more: several A-50 airborne early warning aircraft were lost in the Russia–Ukraine war, its intended successor the A-100 has long failed to enter series production, and although the S-400 is in service it lacks a matching advanced airborne early warning node [4]. Open reporting indicates that the Russian military’s centralized command approval process requires a considerable amount of time, that there is no seamless data interface between the S-400 and short-range air defense systems, and that target information still has to be entered manually [5]. These problems did not arise by accident. They stand in a direct causal relationship to the centralized command and control that the Soviet Union chose at the time.

Counterfactual deduction has a particular use here. The variable of Soviet collapse truncated the comparison between the two paths: we can observe only the complete unfolding of the American path and the abandonment of the Soviet path halfway through. If, taking as constraints the equipment both sides already possessed, the programs then under development, and the publicly disclosed technological

roadmaps around 1991, one deduces the likely evolutionary sequence of this system under a scenario in which the Soviet Union survived, it becomes possible to temporarily remove the variable of “whether the regime persists” and to examine on its own the explanatory power of system design logic. The results of the deduction can then be checked against Russia’s actual process of degradation, which tests the extent to which the original path choices locked in the subsequent trajectory.

## 1.2 Definition of Core Concepts

This study uses four sets of concepts, which must be set out clearly at the outset to prevent conceptual drift.

The first set comprises “A-shoots-B-guides” and “A-locks-B-shoots-C-guides.” Neither is standard terminology, and this study understands them as two levels on the same capability spectrum. “A-shoots-B-guides” denotes the separation of the detecting platform from the launching platform, with the launching side firing on the basis of external information; “A-locks-B-shoots-C-guides” builds on this by separating out midcourse guidance, producing a three-platform division of labor. The threshold for the former is lower; the latter requires the data link to maintain fire-control-quality precision throughout the entire flight.

The second set comprises “engage on remote” and “Cooperative Engagement Capability.” Engage on remote is the U.S. military’s operational-concept formulation; Cooperative Engagement Capability is the technical system that implements it. The distinction between the two is that the former describes a mode of operations while the latter describes an engineering realization [6]. This study follows the formal U.S. terminology when discussing American materials, and uses the “A-shoots-B-guides” spectrum as a uniform frame of reference in the comparative analysis.

The third set comprises “fire-control-quality data” and “track-level data.” This is the study’s most important diagnostic indicator. Fire-control-quality data denotes raw sensor returns that have not been processed locally by the detecting platform; the receiving platform can use them exactly as it would use the output of its own radar, directly supporting weapon guidance. Track-level data denotes information that has already been processed by the detecting side and is transmitted in the form of target position and velocity; its precision and update rate are generally insufficient to support terminal guidance [6]. The difference between the two lies not in bandwidth but in where the information sits along the processing chain.

The fourth set comprises “system design logic.” This study uses the term to denote the two sides’ differing presuppositions about “how information ought to flow”: one holds that sensor nodes should be distributed and fused at the network level, the other that the situational picture should be concentrated at the higher echelon, which then allocates fire uniformly. It is distinct both from the level of technological capability and from the merits of individual equipment parameters.

### 1.3 Research Method

This study adopts an approach combining historical comparison with counterfactual deduction.

The historical comparison component handles two lines of inquiry. The first is reconstruction: building up, in chronological order, the technological conditions both sides already possessed around 1991, including equipment in service, programs under development, and publicly disclosed technological roadmaps. This component rests strictly on published literature and involves no speculation. The second is verification: tracing the actual evolution of the system Russia inherited after 1991, and in particular the problems exposed in the Russia–Ukraine war, as control material against which the deduction can be tested.

The counterfactual component follows the operational norms proposed by Lebow [7]. Counterfactual reasoning is the inescapable underlying logic of causal inference in comparative politics and international relations, as Fearon has systematically argued [8]; the volume edited by Tetlock and Belkin sets out tests for counterfactual thought experiments at three levels — logical, methodological, and psychological — insisting that the alteration to the historical record must be minimal, the causal chain must be traceable, and the theoretical implications must be refutable [9]. Within Chinese academia, the introduction of this method is represented by Jiang Jianzhong’s survey, which notes that in small-N case studies counterfactual inference is approximated mainly by controlling for confounding variables [10]. The present study is a typical small-N case, and its constraint settings draw directly on the norms above.

The test of the deduction’s results is two-directional. On the one hand, one examines whether the sequences and equipment combinations the deduction produces match the directions in which Russian technology actually evolved. On the other, one examines whether the links the deduction judged to be blocked were indeed left unresolved over the long term. If both directions hold, the variable of system design logic has explanatory power; if they do not, the variable must be respecified.

### 1.4 Sources and Limitations

The materials used in this study fall into three categories. The first is scholarly literature, including journal articles, monographs, and research reports from China and abroad. The second is public technical material on equipment, including manufacturers’ technical digests, military program reports, and defense think-tank assessments. The third is news reporting and conflict-observation material, used in the empirical verification section of Chapter 7.

The interpretation of technical material requires a specific note. Parameters such as weapon range and radar detection distance vary widely across sources. This study adopts only data corroborated by two or more independent sources; single-source data are flagged in the text and excluded from the conclusions. As for material bearing on the Russia–Ukraine war, the information released by both belligerents

carries a propaganda component, and this study therefore gives priority to third-party technical post-mortems over the battle reports of either side [5].

Three limitations must be acknowledged. First, the results of counterfactual deduction cannot be directly falsified. What can be done is to make the deduction process itself testable — by setting out, for each deductive conclusion, the grounds on which it rests and the points at which it might be rebutted, so that it provides an interface for challenge rather than becoming an irrefutable conjecture. Second, many details of Soviet programs remain classified to this day; the reconstruction of the 1991 baseline in this study therefore relies on Western assessments and on the limited material subsequently disclosed by the Russian side, and may carry a systematic bias. Third, the comparison in this study is confined to the air and missile defense domain and does not extend to cooperative engagement in other domains such as anti-submarine warfare or land strike.

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## Chapter 2 · Technological Preconditions for Separating Detection from Firepower, and the Comparative Framework

### 2.1 The Threshold for Cross-Platform Fire-Control-Quality Data Sharing

Having one platform supply firing data on behalf of another is difficult not because of communications but because of time.

To intercept an airborne target maneuvering at several hundred meters per second, the entire chain from detection to impact is compressed into a few tens of seconds. Every additional second that data spends traveling along that chain adds several hundred meters of error to the target's position. This imposes three requirements on the data link: sufficient bandwidth, capable of carrying uncompressed radar returns rather than a processed track summary; sufficiently low latency, so that the receiving platform's fire-control computer can treat externally supplied data as though it were its own detection; and strict network-wide time synchronization, without which observations from different platforms cannot be fused within a common temporal frame [6].

Time synchronization is the requirement most easily overlooked. The approach taken by Cooperative Engagement Capability is to have all nodes in the formation share a single time reference, with each platform broadcasting its own raw observations together with a precise timestamp, and each node's fusion processor performing the synthesis locally [6]. The price is high system complexity and expense; the payoff is that the destruction of any single node will not deprive the network as a whole of fire-control precision. The Soviet path was the opposite: a higher-echelon node centrally processes the data reported by the various radar sites and then passes the processed results down to the fire units. Centralized processing reduces the complexity borne by each node, but what is passed down is a track

rather than a return, and the act of passing it down introduces an additional temporal stage.

The divergence between these two technical paths constitutes the physical basis for all the comparisons that follow.

## 2.2 Openness in Weapon Guidance Regimes

The data link settles whether target information can be delivered; the guidance regime settles whether the missile can use it once delivered.

Semi-active radar homing requires the illuminating radar to remain pointed at the target throughout the flight. This means that the launching platform and the illuminating platform can be separated, but the illuminating platform must accept the risk of exposure, and the missile's range is bounded by the reach of the illuminating radar. Active radar homing puts the seeker on the missile, which homes autonomously in the terminal phase and requires only inertial navigation with data-link updates in midcourse. Under this regime, who supplies the midcourse updates ceases to be critical: in principle any node holding the target track can do it.

Active seekers are the precondition for carrying the separation of detection from firepower to greater depth. Around 1991, both the American AIM-120 and the Soviet R-77 were in service or close to it, and the new generation of long-range surface-to-air missiles in both countries likewise adopted a composite regime of inertial navigation plus terminal active or semi-active homing. In other words, at the level of weapons there was no generational gap between the two sides in 1991. The gap lay in whether the data link could deliver midcourse updates to the missile at fire-control quality.

The case of the Standard Missile-6 deserves separate mention. It fitted an active radar seeker onto the Standard Missile-2 airframe while retaining a semi-active mode, and relies on target updates uploaded over the data link in midcourse [2]. It is precisely this design decision — preserving the midcourse interface — that made it the core weapon of Naval Integrated Fire Control-Counter Air: the missile's range far exceeds the radar horizon of the launching ship, and the excess must be supplied by the network.

## 2.3 The Stepwise Progression from “A-Shoots-B-Guides” to “A-Locks-B-Shoots-C-Guides”

Combining the two technological preconditions above yields a capability spectrum, which this study divides into three levels.

The first level is target designation. One platform tells another where it has detected a target; the receiving platform then turns on its radar, searches, locks on itself, and engages independently. This is the most common form of cooperation, both sides possessed it during the Cold War, and this study does not count it as “A-shoots-B-guides.”

The second level is “A-shoots-B-guides,” that is, engage on remote. The launching platform’s radar is off or cannot see the target; the platform computes firing data directly from externally supplied data and fires, the missile’s midcourse phase is continuously corrected by the launching platform or by a third party, and the seeker homes autonomously in the terminal phase. This level requires that the external data reach fire-control quality and that the data link remain connected throughout the missile’s flight.

The third level is “A-locks-B-shoots-C-guides.” Detection and locking, weapon launch, and midcourse guidance are performed by three different platforms, or guidance is shared between the launching platform and a third party. This level places the highest demands on the network: guidance authority must be transferable between nodes, and the loss of any single node must not interrupt control of the missile. The decomposition used in the Chinese-language literature on the three American kill chains follows exactly this four-element scheme — sensor platform, data link, weapon control system, and guided missile [1, 2].

The three levels do not represent a simple escalation in performance but an escalation in dependence on the network. The higher the level, the greater the system’s combat effectiveness — and the greater the functional loss the system suffers once it is degraded. This point recurs in Chapter 7, in the discussion of the degradation of the Russian system.

## **2.4 The Four-Layer Framework and the Measure of Closed-Loop Completeness**

The organizational philosophies of the U.S. and Soviet systems differ so greatly that comparing equipment parameters directly is liable to mislead. This study therefore handles the comparison through a two-tier structure.

The first tier is a vertical four-layer framework: the sensor layer, the data-link layer, the weapon layer, and the command-and-control layer. The sensor layer examines how detection nodes are distributed and what they can detect. The data-link layer examines the grade of information transmitted — fire-control quality or track level. The weapon layer examines guidance regimes and range coverage. The command-and-control layer examines how concentrated decision authority is and how long the decision cycle runs. This four-layer framework grounds the comparison in system composition, allowing two heterogeneous systems to be brought into dialogue on a common set of dimensions.

The second tier is a horizontal measure of closed-loop completeness, using as its criterion whether the three elements “A (locks) — B (shoots) — C (guides)” run through the entire engagement process. A complete loop means that a single engagement, from detection to impact, requires no human intervention and is not interrupted by the loss of a node; an incomplete loop manifests as some stage that must be made good by the platform’s own local capability, or that must be bridged by manual operation.

This measure borrows from the methodological line of Chinese research on system-of-systems confrontation. Yin Xiaojing and colleagues summarize the basic concepts, taxonomy, and assessment methods for system-of-systems contribution rates, observing that war is a confrontation between the combat systems of the two belligerents and that a contribution rate is a measure of the role a combat element plays within its system [11]; Liang Jialin and Xiong Wei survey methods for assessing the contribution of weapon equipment to a system of systems [12]; Zhao Danling and colleagues propose an assessment method based on the combat loop [13]; and Zhang Mingzhi and Ma Li model and simulate the robustness of the OODA loop in system-of-systems confrontation [14]. Hu Xiaofeng's *On the Science of War* provides an integrative theoretical framework for the study of war as a complex system [15], while the volume he co-authored with Yang Jingyu and others, *Simulation Analysis and Experimentation of Complex War Systems*, sets out concrete methods for exploratory simulation [16]. This study does not attempt a quantitative calculation of system-of-systems contribution rates — under deduction scenarios there is no reliable data basis for one — but it adopts the underlying approach: the value of a piece of equipment within a system is judged by its contribution to closed-loop completeness, not by its individual parameters.

Table 1 sets out the correspondence between the four-layer framework and the three-element loop.

**Table 1 · Correspondence between the four-layer framework and the closed-loop elements**

| System layer              | Corresponding loop element     | Core diagnostic indicator                                                                                                |
|---------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Sensor layer              | A (locks)                      | How nodes are distributed; whether the detected data are raw returns or processed tracks                                 |
| Data-link layer           | The A—B—C connection           | Bandwidth, latency, and time-synchronization precision; whether fire-control-quality updates can be sustained throughout |
| Weapon layer              | B (shoots) and C (guides)      | Openness of the guidance regime; whether the midcourse interface can be taken over by a third party                      |
| Command-and-control layer | The decision stage of the loop | Degree of centralization of decision authority; decision cycle from detection to firing                                  |

## Chapter 3 · The 1991 Baseline: The Technological Starting Point of the American Path

### 3.1 Concept Formation and Development of the Cooperative Engagement Capability System

The concept of Cooperative Engagement Capability can be traced back to the early 1970s, when it was proposed by the Johns Hopkins University Applied Physics Laboratory under its original name, “Battle Group Air Defense Coordination” [17]. Its point of departure was a concrete tactical predicament: within a carrier battle group, the radar of every ship is limited by the curvature of the earth, so that detection range against low-altitude sea-skimming targets is far shorter than against high-altitude ones. When a single ship fights alone, by the time it sees the target itself there is often no longer time to react.

The solution was to have all radars in the formation share raw data. In the foundational technical literature on the system, Grant set out how it works: the ships and aircraft of a formation constitute a unified air defense network through networked communications; the radar data of each platform is continuously passed to the other units; each node synthesizes from these data the same composite sensor picture, thereby enabling a ship or an aircraft to attack a target it cannot itself see [6]. This was among the earliest network-centric systems the U.S. Navy put into practical use.

Nearly twenty years elapsed from concept to engineering. The system entered development in 1987, became a formal acquisition program in 1992, and in 2002 received Department of Defense approval to proceed to fleet-wide procurement and deployment [17, 18]. The development cycle was long chiefly because of two engineering problems: network-wide time synchronization, and fusing the returns of multiple heterogeneous radars into a single track without loss of precision.

This technological line had already completed proof of principle and produced engineering prototypes before the Soviet Union dissolved in 1991. That is to say, the United States did not begin building Cooperative Engagement Capability only after winning the Cold War; the confrontation with the Soviet Union itself gave rise to the system. This point is a key premise of the counterfactual deduction that follows: competitive pressure was one of the causes of this technology, not its result.

### 3.2 The 1995 Sea Trials and Initial Operational Capability in 1996

A sea trial in 1995 completed the critical validation: the USS *Dwight D. Eisenhower*, fitted with Cooperative Engagement Capability, together with two Aegis cruisers, an amphibious assault ship, and a P-3 reconnaissance aircraft, were linked into a single whole; radar data from the distant reconnaissance aircraft entered the local fire-control system and successfully guided shipborne weapons [17]. The significance of this trial lay in proving that data from remote sensors could enter the fire-control circuit directly, rather than serving only for situational display.

In 1996 the system achieved initial operational capability with the USS *Dwight D. Eisenhower* carrier battle group [17]. In a live-fire exercise off Hawaii in January of that year, an Aegis cruiser fired Standard Missiles in succession and hit targets beyond the radar horizon of the firing ship itself [19]. This was the point at which “engage on remote” passed from concept to fact.

As of the 1991 cross-section, what the American side already possessed was: a fire-control-quality data link validated at sea; a remote-engagement loop already formed and validated at sea; and a fleet in the course of transitioning to active-seeker missiles. What it did not yet possess was the integration of stealthy sensor nodes and cross-service system integration — neither of which was achieved until the 2010s.

### 3.3 The Three Kill Chains of Naval Integrated Fire Control–Counter Air

Naval Integrated Fire Control–Counter Air was proposed as a concept in 1996 and established as a capability development program in 2002 [2]. Chinese scholars have systematically examined it: Sun Haiwen and colleagues decomposed it into three kill chains — air-to-air, surface-to-air, and surface-to-surface against land — analyzing for each the composition of sensor platforms, the wireless sensor network, the weapon control system, and the actively guided missile [1]; Gong Shangyu and colleagues discussed the development of the system and approaches to countering it [2].

The three kill chains share a single technical skeleton: over-the-horizon detection provided by the E-2D, by F-35 fighters, or by land-based radar; fire-control-quality data transmission carried by Cooperative Engagement Capability; solution computed by the Aegis system or another fire-control center; and interception executed by the Standard Missile-6 [1, 2]. The difference lies only in the particular combination of sensor and weapon platforms.

In August 2013 the system completed its first at-sea demonstration aboard the USS *Chancellorsville*, intercepting a target drone with a Standard Missile-6 [20]; in 2015 the USS *Theodore Roosevelt* carrier strike group made the system’s first operational deployment [21]; and in 2016 the U.S. military integrated the Standard Missile-6 with F-35B sensors into the system and completed a live-fire test [22]. These three steps correspond respectively to the formation of the loop, the deployment of the loop, and the extension of the loop to stealthy nodes.

In the land-based dimension the task falls to the Army’s Integrated Battle Command System (IBCS), which is the key link for interconnection with the naval system [23]. Zhao Fengmin and colleagues analyzed the composition of the operational capability of the naval integrated fire-control system [24]; Zhou Zhen and colleagues produced an annual survey of world air and missile defense development and drew out implications for China’s own development [25]. Among earlier introductory articles, Liu Weiguo and colleagues and Xun Feizheng and colleagues respectively traced the development of the system and set out the composition of integrated air and missile

defense fire control [26, 27]. These works supply the factual basis in equipment terms for the evolutionary chain of the American direction set out in this study.

### **3.4 Why the Air Force Path Stalled**

The position of the U.S. Air Force stands in contrast to that of the Navy. What follows is summarized from public technical material. The F-22 was fitted with advanced sensors and a stealth design, yet its choice of data link long prevented it from sharing fire-control-quality information with other platforms: in order to preserve stealth, the F-22 used a dedicated low-probability-of-intercept data link which in its early form could not interoperate with Link 16, still less connect into the Cooperative Engagement Capability network. The result was that the F-22 became an information island of excellent performance.

This case carries two implications for the present study. First, it shows that the bottleneck for Cooperative Engagement Capability lies not in the performance of any individual platform but in the choice of network interfaces. The F-22's sensor capability was sufficient to sustain a closed loop; what it lacked was a channel through which to hand its data over. Second, it indicates that system design logic is stable across services: the Navy chose open networking, the Air Force chose closed self-preservation, and both choices related to their respective operational scenarios — the Navy must share the situational picture within a formation, while the Air Force must survive alone inside an enemy air defense envelope.

The F-35 changed this situation. It possesses both stealth and a data-link design capable of interoperating across multiple platforms, and the 2016 live-fire integration test with the Standard Missile-6 marked the formal entry of stealthy nodes into the fire-control network [22]. Thereafter Chinese research began to attend to the concept and key technologies of distributed air operations [28], and to the current state, development trends, and operational scenarios of the “loyal wingman” [29] — extensions of the same technological line.

## **Chapter 4 · The 1991 Baseline: Institutional Characteristics of the Soviet Path**

### **4.1 The Air Defense Forces Structure and Centralized Command and Control**

The organizational form of the Soviet air defense system was directly related to the operational scenario it was built to meet. The Air Defense Forces (PVO Strany) constituted a separate service responsible for the defense of the entire Soviet airspace; the scenario for air defense operations was a large-scale, multi-wave, multi-axis air attack, which required firepower disposed in depth and in echelon, with targets allocated by a unified command authority.

That scenario determined that command authority had to be centralized. Faced with several hundred incoming targets, if each fire unit were to assess threats and decide interception sequences for itself, the result would inevitably be repeated interception of high-value targets while other axes went unaddressed. Centralized command solved the problem of fire allocation; the price was that every engagement decision had to pass along the command chain. In the year of the Soviet collapse, Erickson produced a systematic assessment of trends, techniques, and technology in Soviet command, control, and communications, documenting the organizational logic of this centralized system [3].

A centralized system is not in conflict with the separation of detection from firepower. Indeed, centralized command inherently requires such separation: the higher echelon sees the whole picture while the lower echelon executes — that is itself a separation. The conflict arises at the level of data grade: what the higher echelon passes down to the lower must be processed tracks, because the lower echelon neither needs nor is able to process the raw returns of the entire network.

## **4.2 The Operating Mechanism of Automated Command and Control Systems**

The scale of Soviet investment in automated command and control systems was comparable to that of the United States; its direction was entirely different.

The Polyana-D4 series was a brigade-level automated command and control system for air defense missile troops, designed to command mixed groups of air defense missiles under a unified authority. It connected the S-300V, the Buk, the Tor, and other systems into a single command chain, and could be linked simultaneously to a dozen or more objects, including higher command centers, adjacent command centers, and multiple radar information sources [30]. The system discharged functions including the collection and processing of radar information, air situation reporting, automatic target allocation, and coordination of the combat actions of subordinate units [30].

This system entered service in the mid-to-late 1980s and markedly compressed the battlefield situation update cycle of divisional command posts [3, 30]. This was a genuine Soviet achievement in command automation and should not be underestimated. The difficulty is that what it compressed was the situation update cycle, not the engagement decision cycle: target allocation still had to pass through the command post's automatic solution and manual confirmation, and what the fire units received was the result of the allocation, not raw data from which they could compute solutions themselves.

Measured by this study's diagnostic indicators, the Polyana-D4 occupied a position in transition from the first level to the second. It achieved unified command of multiple systems and automatic target allocation, but it did not achieve lateral flow of fire-control-quality data among fire units. Information remained vertical throughout: from radar to command post, from command post to launcher.

### 4.3 The Distributed Firepower Architecture of the S-300 Series

The position of the S-300 series on the Soviet side was equivalent to that of the Aegis system on the American side, but the design premises of the two differed.

The S-300P series was oriented toward homeland air defense, the S-300V series toward army field air defense with an anti-missile role, and the S-300F was the shipborne variant [31]. The firepower architecture of this system has one important feature: launch vehicles may be deployed at a distance from the guidance radar site and controlled remotely by that radar over a radio data link, with the missile's launch and guidance performed by the distant illuminating radar [32]. This solved a concrete terrain problem — the guidance radar must be elevated to obtain low-altitude coverage, whereas launch vehicles must be concealed behind fortifications or terrain masking.

This is already one form of the separation of detection from firepower, but it is not the same thing as the separation on the American path. The S-300's separation occurs within a single system; the two separated parts are under the unified control of one and the same guidance radar, and the function of the data link is to extend that radar's physical control range. The American separation occurs between different platforms: the detecting and the launching parties belong to different ships or even different services, and the function of the network is to make mutually independent systems cooperate.

The former separation is rigid — a closed loop internal to the system — so that once the guidance radar is destroyed, all launch vehicles assigned to it simultaneously lose their combat capability. The latter separation is elastic: the loss of any node can be made good by other nodes in the network. The consequences of this difference for survivability are discussed specifically in Chapter 7.

As to guidance regimes, later models of the S-300P series employed composite guidance combining inertial navigation with terminal active homing or track-via-missile (TVM) operation, in which the missile seeker returns detection information to the ground fire-control radar, the ground station computes the corrections, and commands are then uploaded [31]. Midcourse correction under this regime depends entirely on the system's own guidance radar and cannot be taken over by an external node — and this is precisely the technical reason why it remained at intra-system separation and could not move toward cross-system cooperation.

### 4.4 The Target-Downlink Capability of the A-50 and Its Precision Boundaries

The A-50 airborne early warning aircraft was the piece of Soviet equipment closest to “cross-platform fire-control cooperation.” Based on the Il-76 transport, its prototype first flew at the end of 1978 and it entered service in 1985 [33]. Its airborne radar can detect large airborne targets at ranges of several hundred

kilometers, track upwards of a hundred targets, and guide friendly fighters into contact [33].

The A-50 could pass target coordinates down to ground-based air defense systems over a data link, and could also guide interceptors to the interception. This constituted the de facto prototype of “A-shoots-B-guides” on the Soviet side: the airborne early warning aircraft detects, and ground fire units or interceptors fire.

But this prototype stopped at the first level and did not advance to the second, for three reasons. First, what was passed down was track data rather than radar returns, and its precision and update rate were insufficient to sustain midcourse guidance of a long-range missile. Second, the airborne early warning aircraft and the ground-based air defense systems belonged to different services; their data-link regimes, coordinate systems, and communications procedures were not unified, interconnection required dedicated interface equipment, and the quantity actually fielded was limited. Third, the A-50’s detection capability and target-processing capacity were not themselves weak. Public sources differ on its maximum detection range, though they agree on the order of magnitude; the shortfall lay not in detection but in update rate — its radar is mechanically scanned, with a data update cycle measured in seconds, whereas fire-control-quality cooperation requires an update rate measured in fractions of a second.

Of these three reasons, the first is institutional and the latter two are engineering. Engineering problems can be solved by upgrade; institutional problems require redesigning the way information flows. Before 1991 the Soviet Union did not take the latter step.

#### **4.5 The Attempt at an Interceptor-Based System, and Its Interruption**

Soviet attempts at cooperative engagement in the air-to-air dimension were concentrated on the MiG-31 interceptor system.

The MiG-31 is a heavy interceptor fitted with a high-power phased-array radar, designed to conduct long-range interception within the homeland air defense system. The combination it formed with the A-50 airborne early warning aircraft and ground command posts already possessed, in concept, the prototype of “A-locks-B-shoots-C-guides”: an airborne early warning aircraft or ground radar provides long-range target designation, the interceptor closes with its own radar off, midcourse guidance is provided from the ground or from the airborne early warning aircraft, and locking is completed in the terminal phase by the airborne radar or by the missile seeker. The crux here lies in the seeker regime. A semi-active mode requires the illuminating platform to illuminate the target continuously throughout the missile’s flight; once the airborne early warning aircraft assumes the illumination role, it is in effect pushed to the forward line, and the benefit to the system is cancelled out by the risk of exposure. An active mode allows the airborne early warning aircraft to provide only midcourse updates, with the missile homing autonomously in the terminal phase, and only then can the detecting party and the

illuminating party truly be separated. The difference between these two regimes determined whether this combination could pass beyond the first level.

The associated long-range air-to-air missile, the R-37, entered development in the 1980s, with a maximum range in excess of 200 kilometers — the precise figure varying with engagement geometry and target radar cross-section — and was designed specifically to strike high-value support targets such as airborne early warning aircraft and tankers. The R-37 series existed in both semi-active and active seeker configurations, and the subsequently fielded R-37M employed active radar terminal guidance, which means that at the level of weapons the Soviet Union already possessed the technical conditions for freeing a missile from continuous illumination by its launching platform [34, 35]. The MiG-31M upgrade, fitted with the Zaslon-M phased-array radar of multiplied range and carrying six long-range missiles in tandem in the fuselage bays plus four medium-range missiles under the wings, was the carrier of this system; its first prototype flew in December 1985, and at the end of the 1980s the Sokol aircraft plant built a batch of development prototypes which passed official trials, but the program never entered series production [35, 36].

This line was cancelled in 1994 together with the MiG-31M program [35]. The R-37 missile itself continued to develop after the Soviet collapse, eventually entering inventory in small numbers around 2016–2017 in the form of the R-37M and equipping aircraft such as the MiG-31BM [35]. That is to say, at the level of weapons the Soviet Union (and Russia) retained the capability; what was missing was the system to carry it — the airborne early warning node, the data link, and the command-and-control link all failed to keep pace.

This case is of high value for counterfactual deduction: it sets out the technological reserves that genuinely existed on the Soviet side in 1991, and it marks which links were still missing before those reserves could form a closed loop. The deduction in Chapter 6 takes this as its starting point.

#### **4.6 The Institutional Gap between Track-Level and Fire-Control-Quality Data**

Taking the first five sections of this chapter together, the state of the Soviet side in 1991 can be summarized as follows.

At the sensor layer, the Soviet Union possessed a radar network covering its national territory, airborne early warning platforms, and various field air defense radars; its detection capability was no weaker than that of the United States. At the weapon layer, the range and guidance regimes of the S-300 series and the R-37 missile had reached the advanced level of the time, and active seeker technology had been mastered. At the command-and-control layer, automated command and control systems achieved unified command of multiple systems and automatic target allocation, and command efficiency improved markedly during the 1980s.

The shortfall was concentrated at the data-link layer. Soviet data links transmitted processed track information, not fire-control-quality raw sensor data. This gap cannot be fully explained by insufficient bandwidth or by backward components — it was the necessary product of a centralized system: since the higher echelon processes the situation as a whole, the lower echelon does not need raw returns, and the system's interface standards are consequently not designed to fire-control-quality requirements.

Examining the relationship between the Soviet military-technical revolution and the American revolution in military affairs, Adamsky argues that it was the Soviet Union which first perceived, in the mid-1970s, the impact of information technology on the forms of war, and that the United States later overtook it by developing that perception into a revolution in military affairs [37]. Freedman traces the origins of the revolution in military affairs, noting that Soviet theorists called the impact of nuclear weapons the “military-technical revolution in affairs,” and that by the early 1970s the core technologies — precision guidance, long-range remote control, target recognition and acquisition, command and control — could in fact already be enumerated [38]. Leese, taking the Cold War computer arms race as a case, shows that export controls forced the Soviet Union into a prolonged catching-up position in computers [39].

The relationship between this study's variable and the research above must be made explicit. Adamsky and Freedman deal mainly with a cognitive question: which side first perceived the impact of information technology on the forms of war, and which then developed it into a systematic military doctrine. This study deals with the next link: after both sides had perceived the same set of technological conditions, what kind of information-flow structure did their respective engineering systems ultimately form? The former is a dispute about who perceived first; the latter is a dispute about path dependence. The material on the Soviet side shows that it was not behind at the level of perception — indeed it was earlier — but that in the structure of information flow it chose a solution different from the American one, and it is this choice that the present study seeks to explain. Only by separating the two questions can it be shown that system design logic is not a variable already covered by the existing literature.

These studies settle the question of “who thought of it first,” but say little about “what each side's engineering system ultimately became once it had been thought of.” The point this study adds is that the Soviet Union both thought of it and built a system of its own; the problem lay neither in a failure to conceive nor in a failure to construct, but in the fact that what was constructed made a different choice from the American one about how information should flow, and that choice locked in the level of cooperative engagement that could subsequently be reached.

## Chapter 5 · Comparison of the Two Paths

### 5.1 Data-Link Grade: Fire-Control-Quality Raw Data versus Processed Track Information

The most fundamental divergence between the two paths lies in what the data link transmits.

On the American side, what is transmitted are raw returns. Each node in the Cooperative Engagement Capability system broadcasts its own radar observations together with a precise timestamp to the entire network, and each node's fusion processor synthesizes a composite track locally [6]. Alberts and colleagues' *Network Centric Warfare* is precisely the theoretical expression of this technical line, articulating the mechanism by which information superiority is translated into decision superiority [40].

On the Soviet side, what is transmitted are tracks. The automated command and control system centrally processes the data reported by the various radar sites and passes down target allocation results and track parameters to subordinate fire units [3, 30]. This design is entirely self-consistent with the regime: since the higher echelon is responsible for the overall situational picture, subordinates have no need for raw returns.

The consequences of these two grades of data differ. Fire-control-quality data allows any node at any moment to complete weapon solutions independently; the effectiveness of the network equals the sum of the detection capabilities across the entire network. Under track-level data, the upper bound on a fire unit's combat capability is set by the precision and update rate of the tracks passed down, and the effectiveness of the network equals the upper bound on the processing capacity of the command post. The former is additive; the latter is constrained by the narrowest link.

Chinese research on airborne data-link technology provides a technical account of this difference [41]: the update rate and coordinate precision required for missile midcourse guidance can be stably supplied only by the fusion of raw observations; tracks that have undergone one stage of centralized processing before being passed down accumulate error in transmission, and their update cycle is governed by the processing cadence of the command post.

### 5.2 Misalignment between Closed-Loop Completeness and the Timeline of Realization

Measured against the three-element criterion set out in Chapter 2, the state of closed-loop completeness on each side in 1991 was as follows.

On the American side, the sea-based "A-shoots-B-guides" loop had already been formed, achieving initial operational capability in 1996 [17, 19]. Of the three

elements, A (locks) was borne by any radar within the formation, B (shoots) by an Aegis ship, and C (guides) was still performed by the launching ship's Aegis system and had not yet become an independent node. The American state from 1991 to 1996 was therefore one in which "A locks, B shoots" was complete and "C guides" was partially independent.

On the Soviet side, the land-based direction had achieved intra-system separation of detection from firepower: S-300 launchers could be deployed far from the guidance radar and operated under its remote control [32]; in the air-to-air direction, the A-50 airborne early warning aircraft possessed the capability to guide interceptors [33]; but both directions stalled in the transition zone between the target-designation level and the remote-engagement level, without forming a cross-platform loop at fire-control quality. Of the three elements, A could operate across platforms, while B and C remained bound within the same system.

The temporal relationship between the two paths deserves separate discussion. From concept proposal to loop formation, the United States took roughly twenty-five years, with principle validation completed in 1995; the Soviet automated command and control system entered service in the mid-to-late 1980s, earlier than the formation of the American loop, but it did not progress toward fire-control quality. In other words, the Soviet Union did not move slowly; it moved along a different path, and that path had already reached a considerable degree of maturity within its own logic.

This judgment departs from the mainstream Chinese narrative. Discussion of the Soviet air defense system in China has mostly concentrated on equipment parameters — range and the number of targets that can be engaged simultaneously. Pelosi's analysis of the adversarial relationship between the S-300 and S-400 generations from the perspective of cruise-missile penetration follows the same line [42]. The judgment of this study is that at the architectural level the Soviet system had long possessed the skeleton of detection–firepower separation, and that its weakness lay in data precision and latency across command echelons rather than in the system concept itself. This conclusion is drawn from a synthesis of the material in Chapter 4 and forms the basis for the deduction that follows.

### 5.3 The Roots of the Divergence in System Design Logic

The divergence was not an accidental technical misstep; it had three sources.

The first was the operational scenario. The American scenario was blue-water operations: carrier battle groups operating far from home territory, with no land-based radar network to rely on, so that the formation had to carry its own complete detection capability and horizontal sharing among nodes was a precondition for survival. The Soviet scenario was homeland air defense: behind the forces lay a nationwide radar network and echeloned fire dispositions, detection resources were abundant, and the problem to be solved was how to allocate fire rationally

among thousands of targets, for which centralized command was the most direct answer.

The second was service structure. The U.S. Navy is a highly autonomous service; at sea, command authority over the formation is vested in the formation commander, which provided organizational space for autonomous coordination among nodes. The Soviet Union constituted homeland air defense as a separate service, and command authority over air defense operations was drawn up the vertical chain; horizontal coordination lacked an organizational foothold. The case of the U.S. Air Force confirms this point in reverse: the F-22 chose a closed data link precisely because it was tasked with independent operations deep inside enemy territory, a requirement different from the coordination needs of a naval formation.

The third was the industrial system. Evangelista's comparative study of military technology development in the two countries finds that the combination of competitive pressure, bureaucratic structure, and technological base determines whether a side can translate scientific potential into equipment advantage [43]. Soviet electronics had strengths in analog devices and high-power microwave components, and lagged persistently in digital signal processing and large-scale integration under the impact of export controls [39]. Fire-control-quality data fusion is precisely the engineering task most dependent on the latter: network-wide time synchronization, multi-source return registration, and real-time track synthesis all depend on digital processing capability. The Soviet technological base was not incapable of supporting centralized processing; it was ill-suited to supporting network-wide distributed processing.

In discussing how technological progress shapes the rise and fall of great powers, Ding observes that scholarship habitually focuses on the moment of innovation, whereas a more effective explanation places the emphasis on technological diffusion [44]. This perspective can explain why the Soviet shortfall in fire-control-quality data fusion was not a problem of single-point innovation but one of diffusion: distributed processing requires not only advanced processors but also distributing processing capability to every node and unifying interface standards across the entire network — behind this lies the diffusion of an entire supply chain and organizational capacity. The lock-in effect of interface standards has been studied in civilian communications; Shen Yi and Gong Yunmu summarize it as “technological authority,” that is, winning other states' voluntary followership by locking in the trajectory of technological development and the rules of governance [45]. This mechanism transfers directly to military systems: once a data-link interface standard is established, subsequent equipment can develop only along that trajectory, and switching costs increase over time. Both the Soviet track-level data regime and the American fire-control-quality data regime belong to this class of already-formed trajectories.

In surveying twentieth-century great-power military competition, Li Chen observes that where the demands of military competition and the degree of domestic and international mobilization remain stable over the long term, the development and

application of new technologies become the main engine of competition, and missile weapons enter the combat inventories of all three services in the course of this process [46]. What must be added is that the direction of development is itself shaped by the form of competition. The United States and the Soviet Union faced the same set of technological conditions, but their respective operational scenarios and industrial bases combined those conditions into two different systems.

Table 2 summarizes the comparison of the two paths across the four-layer framework.

**Table 2 · 1991 four-layer comparison of U.S. and Soviet cooperative engagement systems**

| System layer              | American path                                                                                          | Soviet path                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Sensor layer              | Distributed deployment within the formation; nodes carry their own detection capability                | Nationwide radar network plus airborne early warning platforms; detection resources abundant      |
| Data-link layer           | Network-wide broadcast of raw returns; fire-control quality                                            | Tracks processed centrally and passed down; track level                                           |
| Weapon layer              | Active seeker plus midcourse data-link interface; midcourse updates can be taken over by a third party | Active or track-via-missile composite guidance; midcourse dependent on the system's own radar     |
| Command-and-control layer | Formation commander authorizes; nodes compute autonomously                                             | Vertical command chain; higher echelon allocates targets uniformly                                |
| Closed-loop completeness  | A locks and B shoots complete; C guides partially independent                                          | Intra-system separation complete; cross-system cooperation stalls at the target-designation level |

## Chapter 6 · Counterfactual Deduction: System Evolution under a Scenario of Cold War Continuation

### 6.1 Starting Conditions and Three Constraints of the Deduction

The alteration to the historical record is set at 1991: the Soviet Union does not dissolve, and Cold War-style competitive pressure continues to the present day. Following the test criteria set by Tetlock and Belkin, the alteration must be minimal [9]; this study therefore changes only the single variable of “regime persistence,” treating all other conditions according to historical fact, including the industrial bases, technological reserves, geographic environments, and service structures of both sides.

Replacing a political event with a technological–organizational complex spanning thirty-five years as the object of deduction requires additional constraint mechanisms; otherwise the exercise slides easily into unconstrained imagination. This study sets three constraints.

The first is the technological-inheritance-chain constraint. Every deduced piece of equipment must be traceable to a program that existed in 1991 or to an explicit technological roadmap of a design bureau. Introducing new regimes without provenance, or transplanting a mature scheme from one country directly to the other, is not permitted.

The second is the engineering-cycle constraint. Physical laws and development cycles cannot be overridden; competitive pressure can compress schedules but cannot cancel them. The compression adopted in this study takes the actual American experience as its upper bound: Cooperative Engagement Capability took roughly twenty-five years from concept to closed loop; under sustained competitive pressure this is compressed to fifteen to twenty years. No additional relaxation is granted on the Soviet side, because what this study seeks to test is precisely the role of system design logic, not the role of resource investment.

The third is the continuity-of-system-design-philosophy constraint. The United States evolves along the distributed-sensor-network line and the Soviet Union along the centralized-command-and-control line, without unjustified jumps between routes. This is the strongest of the three constraints and also the most open to challenge. Its justification rests on the analysis in Chapter 5: route choice is jointly determined by operational scenario, service structure, and industrial system, none of which changes under the deduction scenario.

Every deductive conclusion states the grounds for the judgment and the points at which it might be rebutted, so as to provide an interface for challenge. This treatment follows the operational norms Lebow designed for counterfactual research in international relations [7].

## **6.2 The American Direction: Network Expansion, Integration of Stealth Nodes, and All-Domain Fusion**

Phase I (1991–2005): network expansion and consolidation of the sea-based loop. The grounds for this judgment are historical fact — Cooperative Engagement Capability became a formal acquisition program in 1992, completed sea trials in 1995, achieved initial operational capability in 1996, and was approved for fleet-wide procurement in 2002 [17, 18, 19]. The only difference under the deduction scenario is procurement scale: sustained competitive pressure would accelerate the pace of fielding, but the technological sequence remains unchanged. At the end of this phase, the sea-based “A-shoots-B-guides” loop covers principal surface combatants, and the C (guides) element is still borne primarily by the launching ship.

Phase II (2005–2015): integration of stealth sensor nodes into the fire-control network. The grounds for this judgment are the development schedule of the F-35

program and the construction timeline of Naval Integrated Fire Control–Counter Air — concept proposed in 1996, established as a capability-building program in 2002 [2]. In historical fact, the first sea demonstration was completed in 2013 and the first operational deployment occurred in 2015 [20, 21]. Under the deduction scenario, the integration of stealth nodes is an inevitable direction: once the adversary possesses a stealthy penetration capability, detection nodes must be pushed forward, and the survivability of forward-deployed nodes can be secured only through stealth. At the end of this phase, the C (guides) element becomes independent and can be borne by an airborne early warning aircraft or a stealth fighter; the three-platform division of labor of “A-locks-B-shoots-C-guides” is established.

Phase III (2015–2026): all-domain fusion and manned–unmanned teaming. The grounds for this judgment are the 2016 live-fire integration test of the F-35B with Standard Missile-6 [22] and the construction progress of the Army’s Integrated Battle Command System [23]. The direction of the deduction is to extend the sea-based loop to land and air, and to incorporate unmanned platforms into the network as expendable sensor nodes. Chinese analyses parsing the concept of Distributed Air Operations and its key technologies [28], and studies of the status, developmental trends, and operational scenarios of the loyal wingman [29], reflect the extension of this technical line in reality; research by Li Hongyu and colleagues on near-space hypersonic weapons and defense systems identifies a new threat driver [47]. At the end of this phase, the loop is no longer confined to a single service, and both the number of nodes and the resilience of the network increase markedly.

The credibility of this deductive line is relatively high because it overlaps heavily with historical fact — the American path was never interrupted by the events of 1991. The value of the deduction lies in using it as a frame of reference against which to calibrate the timescale of the Soviet path.

### **6.3 The Soviet Direction: Extreme Evolution on Land, Breakthrough of the Air-to-Air Loop, and Sea-Based Expansion**

Phase I (1991–2005): extreme evolution of the land-based air defense system. The grounds for this judgment are the technological roadmap of the Polyana-D4 series [30] and the improvement cadence of the S-300 series [31]. The direction of the deduction is not a turn toward distribution but pushing the centralized framework to its limit: raising the automation level of command posts, shortening the target-allocation cycle, expanding the number of fire units a single system can command, and extending intra-system separation of detection from firepower to greater distances.

The technological gains of this phase are predictable: missile range continues to increase, terminal active guidance becomes widespread, and multi-target engagement capability improves. But under the third constraint, no systemic change occurs at the data-link layer — what is passed down remains tracks, not returns.

Closed-loop completeness therefore stays at the level of intra-system separation, and cross-system “A-shoots-B-guides” fails to materialize.

Possible rebuttal: if fire-control-quality data transmission is achievable in engineering terms, why did the Soviet Union not refit? The answer is that refitting would mean distributing processing capability to every fire unit, which conflicts with the organizational logic of the centralized regime; under the homeland air defense scenario, the fire-allocation benefits of centralized processing outweigh the coordination benefits of distributed processing, and the regime lacks internal motivation to refit. This judgment can be challenged; the way to challenge it is to find evidence that the Soviet Union did in fact attempt horizontal fire-control sharing after 1991. No such evidence exists in currently available public materials.

Phase II (2005–2015): breakthrough of the air-to-air “A-shoots-B-guides” loop. The grounds for this judgment are the technological reserves of the MiG-31M and R-37 [35, 36] and the upgrade path of the A-50 airborne early warning aircraft [33]. Two of the necessary conditions already had an engineering foundation in 1991: the ability of the MiG-31 and A-50 to pass target data downward has been confirmed in open reporting, and the active radar terminal guidance of the R-37M allows the missile to dispense with continuous illumination by the launching platform; what was missing was not any single technology but the integration engineering needed to connect the two onto a common data regime. The direction of the deduction is: the airborne early warning radar is replaced with a phased array and its data rate raised, the data link between the airborne early warning aircraft and the interceptor is opened, enabling the interceptor to launch long-range missiles on the basis of airborne early warning data with its own radar off, and the missile is continuously corrected in midcourse by the airborne early warning aircraft.

The key to this deductive line is that it does not require the Soviet Union to change its system philosophy. Airborne-early-warning-guided interception was already a standard mode of operations under the centralized regime; raising the guidance data from track level to fire-control quality is an engineering upgrade within the existing architecture, not a jump between routes. It therefore satisfies the third constraint.

Time estimate: under the engineering-cycle constraint, starting from the technological reserves available in 1991 and taking account of three engineering tasks — replacement of the airborne early warning radar, unification of the data-link regime, and modification of the missile’s midcourse interface — the compressed duration is roughly fifteen to eighteen years, implying technical validation around 2006–2009 and operational capability around 2012–2015. This estimate is notably later than the American sea-based loop (1996) and the American air-to-air loop (live-fire validation of the F-35B node in 2016 [22]), with a gap of roughly ten to fifteen years.

Phase III (2015–2026): triple blockage of sea-based expansion and space-layer integration. The grounds for this judgment are the technological roadmap of the

Soviet shipborne air defense system S-300F [31] and the actual progress of carrier and shipborne airborne early warning construction. The latter was not a blank in 1991: construction had already begun on the Project 1143.7 nuclear-powered aircraft carrier *Ulyanovsk*, designed with catapults, and its companion shipborne airborne early warning solution was the Yak-44E, developed by the Yakovlev design bureau from 1979 onward; by the time of the Soviet collapse the project had entered prototype construction and a full-scale mockup had been built [48]. The direction of the deduction is to transfer the experience gained on land and in the air-to-air domain to the sea-based domain, with shipborne fixed-wing airborne early warning providing beyond-the-horizon detection and shipborne air defense missiles executing remote engagement.

The deductive conclusion for this phase is blockage, and the blockage has three layers, of which electronics industry is only one. The first layer is shipborne digital processing capability: a sea-based loop requires shipborne nodes to possess real-time fusion processing capability equal to that of land-based nodes, and the shipborne environment imposes stricter constraints on device volume, power consumption, and reliability. According to Leese's study of the Cold War computer arms race, export controls kept the Soviet Union in a persistent catch-up position in computing [39]. Extending this constraint to 2026, the Soviet Union (under the deduction scenario) remains roughly one generation behind the United States in shipborne digital processing capability. The second layer is the precision limitation of the shipborne airborne early warning node: at the starting point of the deduction the Yak-44E had entered prototype construction and entry into service according to plan was not in doubt, but it carried a pulse-Doppler radar and faced the same class of precision boundary as the A-50 discussed in Section 4.4 — the design objective of an early warning radar is long-range search and target allocation, not the provision of continuous fire-control-quality data for long-range missiles; raising its output from track level to fire-control quality requires a phased-array replacement cycle comparable to that of land-based airborne early warning, and the constraints on volume, power consumption, and reliability in the shipborne environment make this replacement even harder. In reality the project was cancelled in 1993 for budgetary reasons, and the Soviet (Russian) navy has since been able to perform beyond-the-horizon detection only with airborne early warning helicopters; all fixed-wing proposals have been aborted. The detection range and endurance of those helicopters are markedly inferior to what a fixed-wing platform would provide, so that the outer edge of the formation's air defense umbrella is compressed to the coverage envelope of helicopters — in contrast to the land-based direction, which can draw on a nationwide radar network. It should be noted that retaining the project under the deduction scenario actually strengthens the blockage judgment: at the starting point there was a platform, drawings, and a full-scale mockup, yet the sea-based fire-control-quality loop still could not be crossed; the constraint lay in the regime and the industrial base, not in the existence of a design. The third layer is horizontal data interfaces: according to currently available public materials, midcourse correction in Soviet (Russian) naval shipborne air defense systems relies primarily on the ship's own guidance radar, and the capability to share fire-control-

quality data across ships is limited. This layer is inferred from the absence of public materials, and its evidentiary strength is lower than that of the first two. Integration of the space layer is likewise blocked: early warning satellites can provide early warning, but their data rate is insufficient to supply fire-control-quality information.

Possible rebuttal: could Soviet electronics achieve a breakthrough under sustained competitive pressure? The judgment here is that a breakthrough would require the indigenization of the entire microelectronics supply chain, which exceeds the capacity of any single service or design bureau; Evangelista's research shows that the technological base is one of three conditions determining whether scientific potential can be translated into equipment advantage [43], and the cycle for changing the technological base is far longer than the cycle for developing a single piece of equipment. Even if a breakthrough occurred, it would be difficult for it to manifest in the sea-based loop before 2026.

#### 6.4 Comparison of System Capabilities on Both Sides in 2026

Extending both deductive lines to 2026 yields the comparison shown in Table 3.

**Table 3 · Comparison of system capabilities on both sides in 2026 under the deduction scenario (layer by layer)**

| System layer              | American direction<br>(deduction)                                                                      | Soviet direction (deduction)                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensor layer              | Manned–unmanned hybrid distributed network; stealth nodes routinely deployed forward                   | Land-based radar network plus phased-array airborne early warning; ample nodes but limited forward-deployment capability                          |
| Data-link layer           | All-domain fire-control-quality fusion; cross-service interoperability                                 | Fire-control quality achieved in the land-based and air-to-air directions; sea-based direction remains at track level                             |
| Weapon layer              | Long-range missile range covers all nodes in the network; midcourse can be taken over by a third party | Long-range missile range reaches a comparable order of magnitude; midcourse correction relies on the system's own radar or airborne early warning |
| Command-and-control layer | Nodes compute autonomously; command authority delegated to the formation                               | Command authority remains concentrated at the higher echelon; automation increases but the decision chain is not shortened by an echelon          |
| Closed-loop completeness  | The three elements of “A-locks-B-shoots-C-guides” run through all domains; nodes are interchangeable   | “A-locks-B-shoots-C-guides” runs through the land-based and air-to-air directions; the sea-based direction is unclosed                            |

This result requires careful interpretation. It is not a conclusion of “American victory, Soviet defeat” but a conclusion of “two systems, each with its own zone of advantage in different operational spaces.”

The Soviet direction’s zone of advantage is home territory and adjacent airspace. Within this zone, the land-based radar network provides near-continuous coverage, command posts hold a complete situational picture, fire is disposed in depth across multiple echelons, and any intruder faces layered interception. The efficiency of the centralized regime is highest here because the fire-allocation problem is most acute.

The American direction’s zone of advantage is the open ocean. Within this zone there is no land-based radar to rely on; the formation must carry its own detection and firepower, and the resilience of the distributed network becomes the decisive factor. Once the Soviet system leaves the coverage of the homeland radar network, the information source for its centralized command breaks.

## **6.5 Respective Zones of Advantage and Structural Weaknesses**

Each side’s structural weaknesses are the obverse of its advantages.

The Soviet direction’s weaknesses concentrate on node vulnerability and away-theater capability. The effectiveness of the centralized regime depends on two classes of high-value node — command posts and airborne early warning aircraft — and the combat capability of subordinate fire units drops sharply once these nodes are destroyed. The intra-system separation of the S-300 exacerbates this: loss of the guidance radar causes all assigned launchers to fail simultaneously [32]. In away-theater operations, lacking the support of a land-based radar network, the airborne early warning aircraft becomes the sole high-altitude detection source, and its survivability problem is magnified.

The American direction’s weaknesses concentrate on cost and complexity. Network-wide fire-control-quality fusion requires every node to carry a high-performance fusion processor and to maintain strict time synchronization; equipment and maintenance costs are far higher than under a centralized scheme. The system is highly dependent on the network: once the network is suppressed or deceived, the autonomous combat capability of individual nodes is inferior to that of traditional systems possessing a complete local loop. A further cost lies in the delegation of command authority under the distributed regime, which raises coordination and authorization issues in multinational operations.

The two paths do not stand in a simple relation of better and worse. Centralization has its own rationality in homeland air defense, unified target allocation, and cost containment; distribution excels in node survivability, resistance to interference, and flexibility in fire allocation. The respective conditions of applicability of the two paths are the focus of this chapter.

## Chapter 7 · Path Verification: Degradation and Adjustment of the Russian System after the Soviet Collapse

### 7.1 The Rupture of the Industrial Base

The first step in verification is to establish what actually happened in 1991, so as to distinguish which degradations arose from the interruption of the regime and which arose from system design logic itself.

After the Soviet collapse, the defense sector's share of the national economy fell sharply from its average level of 1988–1990; military expenditure reached a low point in 1998, and funding for program development all but evaporated [49]. The previously highly specialized division of labor fragmented accordingly: critical links such as advanced radar components, aero-engines, and gas turbines for large warships ended up located outside Russia, and the supply chain could not be closed for a considerable period [49]. Zysk examines the difficulties Russia faces in connecting its defense innovation to the fourth industrial revolution, noting that Russian officials themselves acknowledge how hard it is to catch up with the West after twenty years of stagnation [4].

This experience is commonly summarized as the “lost three decades” of the Russian defense industry [4, 49]. It explains why the gap between the deduction scenario and reality is so large: in reality Russia lost not only competitive pressure but also the industrial base needed to keep the system running. Directly comparing the deduction's results with actual trajectories would therefore overstate the role of system design logic, and the factor of industrial rupture must be separated out.

The method of separation is to compare directions rather than levels. Industrial rupture causes a fall in capability levels, but it does not alter the direction of technological choices. If, in reality, Russia under limited resources continued to invest preferentially in centralized command and land-based air defense rather than turning to a distributed fire-control network, then the lock-in effect of system design logic is supported.

### 7.2 The Constraint Imposed on the Whole System by the Missing Airborne Early Warning Node

Several A-50 airborne early warning aircraft were lost in the Russia–Ukraine war, and the intended successor, the A-100, has long failed to enter series production [4, 5]. This fact bears directly on the test of this study's deduction.

According to the deduction in Chapter 6, the air-to-air “A-shoots-B-guides” loop on the Soviet side could not have formed until between 2006 and 2015, and its technological precondition was that the airborne early warning radar be replaced by a phased array with a higher data rate. In reality Russia travelled exactly this road: the A-50 was upgraded to the A-50U, fitted with a new radar and digital processing system, and the A-100 program was launched in the 2000s with the

objective of fielding a new-generation airborne early warning aircraft carrying a dual-band active electronically scanned array radar [4]. The direction matches the deduction.

The point of blockage also matches. The deduction judged that the Soviet direction would be blocked in its third phase in the sea-based and space-based dimensions because of the electronics-industry bottleneck; in reality the A-100 program has stagnated for a long time, and public analyses commonly attribute this to interruption in the supply of high-performance chips and active electronically scanned array components [4]. These are not the same thing — in reality it is a supply interruption caused by sanctions, whereas in the deduction it is prolonged catching-up caused by export controls — but both point to the same structural weakness: the degree of autonomy of the microelectronics supply chain.

The constraint that a missing airborne early warning node imposes on the system is global. Although the S-400 is in service, it lacks a matching advanced airborne early warning node; its long-range interception capability can rely only on land-based radar and cannot function beyond the radar horizon [5, 50]. This is precisely the shortfall of a centralized system made manifest in reality: system effectiveness depends heavily on a small number of high-value nodes, and the loss of a node brings not a linear decline in capability but the wholesale failure of a particular mode of operations.

Giles compares the claimed with the actual capability of Russian ballistic missile defense and finds a marked gap between official statements and the state of equipment [50]; Giles and Boulègue analyze specifically which parts of Russian anti-access/area denial capability have been exaggerated and which underestimated [51]. Dalsjö and colleagues conducted an item-by-item check of such capability in the Baltic Sea direction and found a gap between actual coverage and the claims of the various parties [52]. All three studies point to the same methodological issue: in assessing the capability of the Russian system, equipment parameters and system completeness must be considered separately.

### 7.3 Structural Problems Exposed in the Russia–Ukraine War

Taking the war in Ukraine as the testing ground, Dalsjö and colleagues conducted a fairly severe post-mortem of actual Russian combat capability [5]. Taken together with public reporting, three categories of structural problem can be identified.

The first is command latency. The Russian military's centralized command approval process requires on the order of minutes, whereas engaging time-sensitive targets requires a response measured in seconds. This is the inherent cost of a centralized system, and it is of a piece with the situation in the Soviet period, when automated command and control systems compressed the situation update cycle but failed to compress the engagement decision cycle (see Section 4.2).

The second is the discontinuity of interfaces. There is no seamless data interface between the S-400 and short-range air defense systems, and target information still

has to be entered manually [5]. This means that in the three decades and more since 1991, the Russian military has still not solved the problem of lateral data continuity between different systems. The deduction judged that no institutional change would occur at the data-link layer on the Soviet side; this judgment is supported.

The third is cost inversion. Using long-range air defense missiles to intercept low-cost unmanned aerial vehicles and loitering munitions is economically unsustainable. Under a centralized system, fire allocation is decided uniformly from above, making it difficult to select interception means dynamically according to target value; a distributed system permits the use of a nearby node whose cost is proportionate to the target. This problem was not salient in the Cold War period, because the scenario then was large-scale air attack rather than low-cost saturation.

All three categories point to system design logic rather than to the level of equipment. In range and multi-target capability the Russian S-400 is not backward; the problem is that it has been placed within an information-flow structure incapable of realizing its potential.

## 7.4 Two-Directional Cross-Check of the Counterfactual Deduction against Actual Trajectories

Setting the deductive conclusions against actual trajectories item by item yields the results shown in Table 4.

**Table 4 · Two-directional cross-check of deductive conclusions against actual trajectories**

| Deductive conclusion                                                                                                                                   | Actual trajectory                                                                                     | Match?                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| The Soviet direction invests preferentially in land-based air defense and centralized command, and does not turn to a distributed fire-control network | The S-400 is fielded and continuously improved; the command system remains vertically centralized     | Match                                                     |
| An air-to-air fire-control-quality loop requires replacement of the airborne early warning radar, at a date around 2006–2015                           | Both the A-50U upgrade and the A-100 program were advanced during this period, but no loop was formed | Direction matches; timing disturbed by industrial rupture |
| The sea-based and space-based dimensions are blocked by the electronics-industry bottleneck                                                            | The A-100 has long stagnated; shipborne airborne early warning capability has not been established    | Match                                                     |
| System effectiveness depends heavily on a small number of nodes such as airborne early warning                                                         | After A-50 losses, long-range interception capability was markedly                                    | Match                                                     |

| Deductive conclusion                                                                    | Actual trajectory                                                                                                      | Match? |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------|
| aircraft, and node loss causes wholesale failure of a mode of operations                | constrained                                                                                                            |        |
| Lateral data continuity between different systems remains unresolved over the long term | Interface discontinuity persists between the S-400 and short-range systems, and target information is entered manually | Match  |

Of the five comparisons, four match, and one matches in direction but has its timing disturbed by industrial rupture. This result supports the study's central judgment: system design logic is an independent variable with explanatory power, which had already taken shape in the path choices made before 1991 and which continued to operate in the technological evolution after 1991.

The limits of this test must be stated. It is a retrospective comparison, not a predictive test; and because the constraint settings of the deduction themselves drew on publicly disclosed Soviet technological roadmaps, there is a risk of circular reasoning. The way to reduce that risk is to attend to those links which the deduction judged to be blocked, since a judgment of blockage is more easily falsified than a judgment of success — blockage means excluding all of the many possible technological breakthroughs. Two of the five comparisons are judgments of blockage (the sea-based and space-based dimensions, and lateral data continuity), and both were borne out in reality. This handling nonetheless leaves one unresolved alternative explanation: did the blockage arise from the lock-in of system philosophy, or from the rupture of resources after the collapse? The two can be distinguished by a single criterion. If a link failed to be breached during a period of relatively ample resources, the lock-in explanation is the stronger; if stagnation occurred only after resources were severed, the two explanations cannot be separated. Applying this criterion, of the five comparisons the item on the sea-based and space-based dimensions belongs to the former class: the Soviet Union already possessed the relevant technological reserves before the collapse, yet the sea-based loop was not advanced under the resource conditions of the late 1980s, indicating that the constraint lay not in money but in the system. The item on lateral data continuity belongs to the latter class: the Russian military's stagnation of three decades and more coincided with industrial rupture, so that item is compatible with both explanations and cannot on its own support this study's variable. This distinction is set out explicitly so that the strength of the test is not overstated.

## Chapter 8 · Conclusions

### 8.1 The Influence of System Design Logic on the Outcome of Technological Competition

The first conclusion is that explaining the divergence in U.S. and Soviet cooperative engagement paths requires adding a variable beyond the level of technology.

Current discussion of the U.S.–Soviet military-technological gap follows a mainstream narrative of technological generation gap: the Soviet Union was backward in electronics and information technology, and therefore lost. That account can explain part of the facts, and this study does not deny the role of the microelectronics supply chain (Sections 5.3 and 7.2 both bear on it). But it cannot explain why the Soviet Union went down a different road in its choice of system architecture, nor why Russia, even with limited resources, continued to invest along the centralized line after 1991.

The variable this study proposes — “system design logic,” that is, distributed sensor network versus centralized command and control — attends not to whose technology was more advanced but to the two sides’ differing presuppositions about how information ought to flow. This presupposition is shaped jointly by operational scenario, service structure, and industrial system; once formed, it is strongly path-dependent. It determines the standards according to which system interfaces are designed, the level at which processing capacity is placed, and consequently the directions in which later upgrades are easy and those in which they are hard.

The Soviet system conception of 1991 was not backward; within its own logic it was already fairly mature. What was backward was the level it could not reach within that logic: cross-platform fire-control-quality data fusion. As argued in Section 5.2, this judgment diverges from the mainstream Chinese narrative, and its basis lies not in a comparison of equipment parameters but in the reconstruction of the two sides’ information-flow structures carried out in Chapters 4 and 5.

### 8.2 Implications for the Offense–Defense Balance and Strategic Stability

The study’s second conclusion concerns a counter-intuitive proposition: whether the high development of cooperative engagement technology in fact strengthens strategic stability.

According to the deduction in Chapter 6, what the two sides form by 2026 is a configuration in which each prevails at home and is constrained away: in the Soviet direction, multi-layered defense in depth over national territory and adjacent airspace; in the American direction, a complete distributed loop over the open ocean; and, for either side, systemic disadvantage upon entering the other’s area of advantage. Under this configuration the payoff structure of preemption changes. The payoff to preemption depends on the destruction rate of the first strike, and the first-strike destruction rate is limited away from home by two conditions:

incomplete detection coverage means the target list must have gaps, and command latency means the strike window is hard to compress. When these two limits press the first-strike destruction rate below the level at which the adversary retains a second-strike capability, preemption ceases to hold in rational calculation, because the retaliation the striking side must bear exceeds what it can avoid. The configuration deduced in this study falls precisely within this interval: both sides possess complete detection and command support at home, and both consequently fail to obtain a decisive first-strike advantage away from home. The more thoroughly cooperative engagement capability deepens home defense, the lower the relative payoff to away strikes becomes.

Two qualifications must be attached to this chain of transmission. First, it depends on both sides possessing a reliable second-strike capability; if one side loses the means of retaliation after a first strike, the inhibiting effect lapses. Second, it depends on both sides judging each other's systemic disadvantages with reasonable accuracy; misjudging the cost of operating away from home leads one side to overestimate the payoff of a first strike. Both conditions held broadly within the context of the nuclear equilibrium of the late Cold War, but they cannot be generalized to all situations.

This inference must be tested within the lineage of offense–defense balance theory. Lieber argues that technology does not by itself determine the offense–defense balance, and that the mediating roles of geography and military doctrine are equally critical [53]. The deduction results of this study are consistent with that judgment: the effect of cooperative engagement technology depends on the geographical and doctrinal environment into which it is placed. In an open-ocean environment it strengthens the offensive side's power projection; in a homeland air defense environment it strengthens the defender's capacity for denial. Blagden introduces technology and geography as two mediating variables and examines under what conditions competition escalates into conflict [54]; his framework can be applied directly to explain the configuration deduced in this study.

Security dilemma theory offers another angle of test. Glaser revisits the concept of the security dilemma, distinguishing the pure security dilemma from spirals caused by misperception and greedy motives [55]; Tang traces the concept's intellectual history to clarify the strict boundaries of its definition and to dispel the loose generalization with which it has been used in the literature [56]; and Liff and Ikenberry apply this framework to military competition in the Asia-Pacific, testing whether the dynamics of a security dilemma are emerging in that region [57]. By the strict definition, the configuration deduced in this study does constitute a security dilemma: area-denial capability built by one side for defensive purposes is seen by the other as weakening the reliability of its power projection, thereby inducing counter-investment. Whether such a spiral escalates into conflict, however, depends on whether the mediating variable of geography remains stable.

Three conditions govern the validity of the proposition. First, the speed of technological diffusion on both sides must be broadly symmetric; if one side leads

markedly, the boundaries of the areas of advantage shift and the stabilizing effect disappears. Second, the boundaries of the geographical areas of advantage must be clear, with both sides sharing an understanding of “what counts as home”; where the boundaries are blurred, the risk of misperception rises. Third, crisis communication channels must not have been undermined by the compression effect of the technology — cooperative engagement capability shortens the decision cycle, and if crisis management mechanisms do not keep pace, the compressed decision time itself becomes a source of instability. Should any one of these three conditions loosen, the conclusion changes; this study therefore advances it as a conditional inference rather than a general law.

The current level of tension in the international security environment is rising, which makes the stability of these conditions a live problem. In his review of the world nuclear situation, Fan Jishe observes that intensified great-power strategic competition, frequent regional security contradictions, and mounting structural pressure on the existing international nuclear order are compounded by the interweaving of multiple difficulties — spillover of nuclear proliferation, declining efficacy of arms control mechanisms, and weakening strategic stability [58]. Competition at the conventional level is likewise accelerating. In analyzing the development trends of the U.S. missile defense system against the background of great-power competition, Yuan Rongliang finds that changes in threat perception directly drive adjustments in interception layers and technological routes [59]. Understanding how this driving mechanism operates over a competition lasting more than three decades helps in judging the time windows and resource allocations of current system-building programs.

Emerging technologies may alter this configuration. Williams discusses scenarios for asymmetric arms control directed at hypersonic glide vehicles, arguing that such weapons may weaken existing frameworks of strategic stability [60]; Enemark examines from the standpoint of military ethics the combination of the “loyal wingman” concept with artificial intelligence and cognitive enhancement technologies, and issues a warning about cyborg-drone warfare [61]. The deduction in this study stops at 2026 and does not incorporate these two categories of variable, which is one of the boundaries of the research.

### **8.3 The Applicability and Limits of Counterfactual Method in Equipment-System Research**

The study’s third conclusion concerns method. Counterfactual deduction can be extended from the history of political events to the history of equipment systems, but at an additional methodological cost.

Existing counterfactual research has mostly focused on decision nodes: whether a given crisis would have been averted, whether a given policy would have been adopted. The volume *Unmaking the West*, edited by Tetlock, Parker, and Lebow, applies this method to the debate over the inevitability of the rise of the West [62]; Henne discusses how to disentangle competing explanations in qualitative

international relations research [63]; and the most recent application is Kaarbo and colleagues' use of counterfactual method to assess the influence of leader personality on the Western response to the war in Ukraine [64]. The objects of these studies are all short-duration, single-node events.

Replacing the object of deduction with a technological–organizational complex spanning thirty-five years magnifies arbitrariness as the time scale lengthens: every link could have taken a different course, and the cumulative space of outcomes becomes too large to constrain. This study's three constraints — technological inheritance chain, engineering cycles, and continuity of system design philosophy — are the treatment of this problem. The third is the strongest and the most open to question; the way of meeting that objection is to ground it in the three-source analysis of Chapter 5 (operational scenario, service structure, industrial system) and to set out explicitly how it might be refuted.

The test of the deduction proceeds by two-directional cross-check, with particular attention to those links judged to be blocked. The reason this method is effective in the present case is that the Soviet collapse supplies an unusually clear body of control material: an inheritor whose regime was interrupted, evolving along the original logic for more than three decades. Such control conditions do not necessarily exist in other cases, so the generalization of the method requires caution.

The probative force of the five comparisons is not equal and must be considered separately. Item 1 (preferential investment in land-based air defense and centralized command) and item 5 (lateral data continuity unresolved) are low-information predictions: any state inheriting the Soviet system and constrained in resources would make the same choices, so their match cannot independently support the variable of system design logic. Item 2 (the timing of an air-to-air loop) and item 3 (blockage in the sea-based and space-based dimensions) are high-information predictions: they concern whether specific links could cross the fire-control-quality threshold, and there is a definite possibility of failure — Russia might have completed lateral continuity or a sea-based loop in the 1990s or 2000s and did not. Item 4 (continued improvement in the land-based dimension) lies between the two. The conclusions of this study rest principally on the two high-information items; the two low-information items serve only as corroborating consistency and bear no argumentative weight.

The Chinese translation of Joseph S. Nye Jr.'s *Understanding International Conflict: From Theory to History* has been influential in China. Its discussion of questions such as “had Bismarck not been dismissed in 1890, would the First World War still have broken out?” provides a popular and forceful defense of the legitimacy of counterfactual reasoning [65]. Beyond legitimacy, this study stresses operability, whose core is to give deductive conclusions an interface at which they can be challenged.

## 8.4 Implications for the Building of Contemporary Cooperative Engagement Systems

The diagnostic criteria proposed in this study can be used directly to assess the cooperative engagement capability of any military force: does the data link carry fire-control-quality data or track-level data; is the loop complete; does terminal guidance require continuous illumination by a third party. Measured by these three indicators, one can avoid being misled by surface indicators such as weapon range.

As to the trade-offs in system building, centralized and distributed architectures each have their conditions of applicability, and the choice should be determined by operational scenario rather than by technological fashion. For a system charged with homeland air defense, the problem of fire allocation is prominent and the returns to centralization are clear; for a system charged with open-ocean operations, the problem of node survivability is prominent and the returns to distribution are clear. Where the two are combined, the design of interface standards matters more than the choice of route — the case of the F-22 shows that a closed data-link choice can prevent equipment of excellent performance from being integrated into a system for a long time.

As to capability assessment, one caution is needed: in assessing an adversary, equipment parameters and system completeness should be considered separately. The Russian case shows that equipment leading in range and multi-target capability, once placed within an information-flow structure marked by interface discontinuity and long command latency, suffers a substantial discount in actual effectiveness [5, 50, 51].

The limitations of this study were set out in Section 1.4 and are summarized again here: the results of the deduction cannot be directly falsified and can be tested only indirectly, through the reasonableness of the constraint settings and the degree of match in the two-directional cross-check; gaps exist in the material on Soviet programs; and the scope of comparison is confined to air and missile defense. Subsequent research could apply the same framework to the anti-submarine or land-strike dimensions, to test whether the explanatory power of the system design logic variable is stable across different operational domains.

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